

FORECASTING ITC STOCK PRICES: A COMPARATIVE STUDY OF ARIMA, SARIMA, ANN, AND LSTM MODELS

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ABSTRACT

Financial decision-making relies heavily on stock price forecasting, which helps institutions and investors control risk and maximize profits. Four popular models are compared in this study for their ability to predict the stock prices of ITC Limited, a diverse Indian conglomerate: Autoregressive Integrated Moving Average (ARIMA), Seasonal ARIMA (SARIMA), Artificial Neural Networks (ANN), and Long Short-Term Memory (LSTM) networks. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE) were the metrics used to train, validate, and assess each model using historical daily stock price data from January 2010 to June 2025. In understanding the fluctuating, non-linear behaviour of financial markets, the results show that machine learning models perform noticeably better than conventional statistical techniques. While ARIMA and SARIMA had limited predicting capacity, especially in volatile settings, ANN demonstrated the greatest performance across all criteria, followed by LSTM. Unexpectedly, the hybrid ARIMA+LSTM model produced the worst outcomes, most likely as a result of overfitting or misalignment in the model integration. This paper illustrates the increased efficacy of deep learning techniques while highlighting the shortcomings of linear models in handling contemporary financial datasets. The findings provide traders, academics, and financial analysts with useful advice on how to choose the best models for stock prediction. Future research on hybrid frameworks, the integration of external factors, and wider sectoral applicability is also encouraged. To sum up, ANN and LSTM offer reliable, flexible, and precise instruments for predicting changes in equity under intricate and unpredictable market conditions.

Keywords: Stock Forecasting, ARIMA, SARIMA, ANN, LSTM, Machine Learning.

INTRODUCTION

Stock price forecasting is one of the most difficult and fascinating areas of financial study because of the erratic and sometimes unforeseen character of financial markets. Since they enable traders, investors, financial institutions, and regulators to make data-driven choices that can reduce risk and increase profits, accurate and timely forecasts are essential. With holdings in FMCG, hotels, paperboards, and agriculture, ITC Limited (Indian Tobacco Company) is a prominent participant among the many businesses listed on the Indian stock exchange. ITC's stock price volatility and financial performance have continuously drawn the interest of institutional and individual investors. As a result, creating accurate forecasting models for ITC's stock prices might give market players important information. A thorough comprehension of the intricate patterns that are impacted by a broad range of political, economic, and psychological issues is necessary for accurate stock price forecasting. Because of their statistical robustness and capacity to represent linear trends in data, traditional time series models—like the Autoregressive Integrated Moving Average (ARIMA)—have long been employed in financial forecasting. Under the presumption of data stationarity, ARIMA models forecast future stock prices using residual errors and past values. Although these models have proven useful in a variety of situations, they are not very good at handling non-linear and non-stationary trends, which are features that are frequently present in actual stock markets.

These constraints may be addressed by the new predictive modeling approaches that have surfaced with the development of artificial intelligence (AI) and machine learning. Significant promise has been demonstrated by Artificial Neural Networks (ANNs) in particular for identifying intricate, non-linear correlations in data. ANNs, which are modeled after the human brain, use training to learn from past data and find patterns that traditional models would overlook. Financial forecasting may benefit greatly from their capacity to simulate chaotic and non-linear processes without assuming anything about the distribution of the data. In order to forecast the stock prices of ITC Limited, this study compares the ARIMA, SARIMA, ANN, and Long Short-Term Memory (LSTM) models. By assessing each model's performance using important metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), the main objective is to determine how accurate and resilient the predictions are. This study will examine the benefits and drawbacks of each model by using it on historical ITC stock data, assisting in the identification of the best approach for various forecasting situations.

This research is important because it makes two contributions: it improves our knowledge of machine learning and statistical models in terms of financial forecasting, and it provides useful information on the best modeling strategy for predicting a stock's performance of big, diverse companies like ITC. In addition to influencing scholarly research in the fields of finance and artificial intelligence, the study's conclusions will offer insightful information for algorithmic trading models and portfolio structuring techniques. In addition to presenting the forecasting results, this article will examine pertinent literature, examine the modeling and assessment procedures, and provide important insights and suggestions for further study.

BACKGROUND OF THE STUDY

A fundamental component of financial research, stock price forecasting provides priceless information to market players of all sizes, from individual investors to major institutional traders. A key component of reducing risks and maximizing rewards is having a high degree of accuracy in future stock price predictions. However, because of a variety of factors, including changes in the economy, political developments, and investor emotion, the financial markets are complicated and frequently display dynamic, non-linear behaviour. Because of these difficulties, predicting stock prices is a challenging and exciting endeavour for both practitioners and scholars.

A strong argument for stock price forecasting is made by ITC Limited, a major participant in India's FMCG, hotel, paperboard, and agricultural industries. Due to its diverse business portfolio and strong market position in India, ITC's stock is highly volatile and susceptible to a number of variables, including shifts in global market dynamics, domestic economic policies, and consumer trends. As a result, both institutional investors and individual traders have expressed interest in forecasting ITC's stock prices. These traders aim to profit from changes in stock prices while minimizing related risks. Over the years, classic statistical methods have been the mainstay of stock price forecasting; one of the most popular approaches is the Autoregressive Integrated Moving Average (ARIMA) model. ARIMA models and forecasts linear patterns using past stock price data. Although ARIMA works well under steady market situations, it has a limited ability to capture intricate, non-linear market characteristics. The chaotic, non-stationary conditions in which financial markets frequently function make it difficult for ARIMA to produce reliable forecasts.

In order to overcome these constraints, scholars have been progressively using more sophisticated machine learning models, especially Long Short-Term Memory (LSTM)

networks and Artificial Neural Networks (ANNs). Artificial Neural Networks (ANNs), which draw inspiration from the structure of the human brain, are able to learn and identify intricate, hidden patterns in historical data without requiring explicit assumptions about the distribution of the data. A specific kind of recurrent neural network (RNN) called an LSTM is excellent at identifying long-term relationships in sequential data, which makes it ideal for time series forecasting, such as stock price forecasting. Since machine learning algorithms can adjust to non-linear patterns and provide more accurate predictions under turbulent market situations, the move towards these models represents a paradigm shift in financial forecasting. The capacity of LSTM to learn from long-term data sequences has drawn particular interest since it is crucial for forecasting the frequently erratic character of financial markets.

The usefulness of machine learning techniques in comparison to more conventional time series models like ARIMA and SARIMA is still up for dispute, despite the increased interest in these methods. Many financial analysts still use ARIMA, although in some forecasting scenarios, machine learning models like ANN and LSTM have demonstrated a strong potential to beat conventional techniques. In order to give a thorough grasp of the advantages and disadvantages of each model as well as its applicability to actual financial applications, this study compares the effectiveness of the ARIMA, SARIMA, ANN, and LSTM models in predicting ITC stock prices. Given this context, the paper tackles important concerns such: How do machine learning models like ANN and LSTM compare to ARIMA and SARIMA models? Which model forecasts ITC stock prices the most accurately? What aspects affect these models' accuracy, and how may they be improved for financial forecasting in the real world? The goal of this comparison research is to provide insight into the best methods for predicting the stock prices of big, diverse firms like ITC in the complicated financial environment of today.

LITERATURE REVIEW

With a variety of approaches used to address the intricacies of financial markets, stock price prediction has long been a major challenge in financial research. In their comprehensive analysis, *Stock Price Prediction using Deep Learning: A review*, Zhang et al. (2020) highlight how deep learning models, such as LSTM, have the ability to identify non-linear correlations in stock price data, something that conventional models, such as ARIMA, frequently fall short of. According to the study, LSTM models often beat ARIMA, particularly under erratic market conditions, demonstrating their capacity to capture intricate patterns and raise forecasting precision. Similarly, in their paper *A Comparative Analysis of ARIMA and LSTM for Stock Market Forecasting*, Kumar, and Gupta (2021) discovered that LSTM performed better than

ARIMA in predicting stock prices, especially during periods of market volatility when LSTM's ability to capture non-linearities became essential. Li et al. (2020) compared SARIMA and LSTM in Time Series Forecasting using SARIMA and LSTM: An Application to Financial Data. They found that LSTM performed better at modeling non-linear trends, but SARIMA was better at capturing seasonality. This highlights the necessity of combining both models for improved forecasting. Chen et al. (2021) analysed different ANN designs, including feedforward neural networks (FNN) and LSTM, and contrasted them with conventional models in Artificial Neural Networks and Stock Price Forecasting: A Review of Approaches. Because LSTM and other ANNs can handle complicated data and uncover hidden patterns in stock prices, they fared noticeably better than ARIMA, according to their assessment.

Singh and Arora's (2022) study, Predicting Stock Market Trends using ARIMA and Machine Learning Techniques, expanded on this analysis by contrasting ARIMA with machine learning models such as SVM and decision trees. Their results showed that machine learning models outperform ARIMA, which had trouble predicting non-linear trends, in predicting stock market patterns, particularly when conditions are turbulent. In their research paper A Comparative article of SARIMA and LSTM for Financial Time Series, Zhang, and Li (2022) Forecasting shown that LSTM was better at identifying long-term dependencies in stock prices than SARIMA, which was stronger at making short-term forecasts but less successful at identifying intricate patterns. In Forecasting Stock Prices: A Review of Machine Learning Models with Focus on LSTM, Gupta et al. (2023) noted that LSTM has emerged as a leading tool for stock price forecasting, surpassing conventional ARIMA and support vector regression, in particular because of its ability to handle non-linearity and long-term dependencies. In The Role of ARIMA and SARIMA in Stock Price Forecasting, Kumar, and Sharma (2023) also discovered that, although ARIMA works well for short-term, linear forecasts, it performs poorly in turbulent markets. In these situations, SARIMA and LSTM perform better, particularly when modeling non-linear trends.

The potential of LSTM in conjunction with technical indicators such as RSI and moving averages was further investigated by Wu et al. (2024) in LSTM-based Stock Price Prediction utilizing Historical and Technical Indicators. The effectiveness of combining numerous data characteristics was demonstrated when they discovered that LSTM, when enhanced with technical indicators, performed noticeably better than both independent LSTM and ARIMA models. In Comparison of ARIMA, SARIMA, and LSTM for Financial Forecasting, Zhang et al. (2024) confirmed similar results, demonstrating that SARIMA outperformed LSTM models

in stable market situations while LSTM models produced the most accurate forecasts in volatile ones. In Stock Price Forecasting with SARIMA and LSTM (2023), Sharma and Kapoor highlighted the necessity of hybrid models that incorporate LSTM for long-term forecasting and SARIMA for short-term prediction. LSTM performs better when anticipating complicated, non-linear market behaviour. According to Reddy et al. (2022) in Hybrid Models for Stock Price Forecasting: Combining SARIMA and LSTM, hybrid models that combine the superior linear trend prediction capabilities of SARIMA with the non-linearity modeling capabilities of LSTM provide a more reliable stock price forecasting solution. The performance of LSTM, ANN, and Random Forest was examined by Patel and Sharma (2022) in Evaluating Machine Learning Techniques for Stock Price Forecasting. They came to the conclusion that LSTM consistently outperformed other machine learning models, especially when forecasting volatile stock prices. Roy et al. (2023) investigated hybrid methods that combined ARIMA with machine learning techniques in Enhancing ARIMA Forecasting with Machine Learning Models for Financial Data. They demonstrated that these models greatly increased forecasting accuracy, particularly in volatile market situations. For long-term forecasting, Gupta, and Mehta (2024) compared LSTM with ARIMA in Long-Term Stock Price Forecasting using LSTM: A Comparative Analysis. They found that LSTM produced better results because it could identify complicated patterns in the data and long-term relationships. Choi et al. (2022) supported similar conclusions in Stock Price Prediction using ARIMA and LSTM: A Comparative Study, demonstrating that ARIMA was unable to predict non-linear price behaviour, whereas LSTM models fared better by capturing both short-term and long-term dependencies.

The benefits of LSTM in managing volatile market data and capturing non-linear connections were highlighted in Kumar and Mishra's (2023) assessment of deep learning models in Deep Learning Models in Stock Market Forecasting. Their research revealed that LSTM outperformed conventional ARIMA models in terms of forecasting, especially in intricate and dynamic market situations. According to Singh et al. (2023) in A Hybrid Model for Stock Price Prediction using SARIMA and LSTM, combining the SARIMA and LSTM models may perform better than using them alone, especially in turbulent financial markets. LSTM surpassed ARIMA in terms of forecast accuracy, particularly for long-term stock price patterns, but ANN also did well for short-term predictions, according to Kumar et al. (2022) in Stock Market forecast: A Comparative Analysis of ARIMA, ANN, and LSTM Models. Last but not least, Soni et al. (2024) in An Evaluation of ARIMA and LSTM Models for Stock Price

Forecasting came to the conclusion that LSTM models continuously beat ARIMA, particularly when predicting intricate, non-linear stock price movements. This finding supports the growing use of machine learning models in stock forecasting.

RESEARCH PROBLEM

Predicting stock prices is still one of the most difficult and important aspects of financial study. Traditional statistical models, such as the Autoregressive Integrated Moving Average (ARIMA), find it difficult to predict stock prices in real-world scenarios due to the extremely volatile and non-linear characteristics of financial markets. The non-stationary and non-linear patterns frequently seen in financial markets are difficult for ARIMA to represent, despite its effectiveness in modeling linear trends (Li et al., 2020; Kumar & Gupta, 2021). However, machine learning models—in particular, Artificial Neural Networks (ANNs) and Long Short-Term Memory (LSTM) networks—have become more popular because of their capacity to manage massive, dynamic datasets and model intricate, non-linear interactions (Chen et al., 2021; Reddy et al., 2022).

Yet, there is still a difference in how well these cutting-edge methods perform when compared to more conventional models like ARIMA and SARIMA, even with the increasing success of LSTM and other machine learning models in stock price forecasting. Few studies have thoroughly compared the performance of ARIMA, SARIMA, ANN, and LSTM when it comes to forecasting the stock prices of particular companies, like ITC Limited, which is involved in several different industries and has a high degree of stock price volatility (Zhang et al., 2020; Kumar & Sharma, 2023). Additionally, the hybridization of SARIMA and LSTM for stock price forecasting in real-world settings has not been well evaluated in research, despite the promise of hybrid models that combine classical and machine learning approaches (Gupta & Mehta, 2024; Zhang & Li, 2022).

Thus, assessing and contrasting the forecasting accuracy and robustness of the ANN, LSTM, ARIMA, and SARIMA models for predicting the stock prices of ITC Limited is the current study subject. The purpose of the study is to determine which model produces the most accurate predictions, especially when it comes to identifying long-term relationships and non-linear market behaviours. Additionally, the study will examine whether hybrid models—which blend machine learning and conventional statistical methods—can enhance predicting performance in turbulent market situations (Singh et al., 2023; Wu et al., 2024). For investors and financial

experts looking for trustworthy resources for forecasting stock movements and making wise investment choices, this issue is essential.

OBJECTIVES

This study's main goal is to assess and contrast how well-advanced machine learning techniques—specifically, Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks—and conventional statistical models—like ARIMA and SARIMA—predict the stock prices of ITC Limited. In doing so, the study seeks to ascertain which model provides the best accurate forecasts, especially when it comes to capturing both linear and non-linear changes in stock prices. In order to capitalize on the advantages of both conventional and contemporary methods in financial forecasting, this study also aims to investigate the possible advantages of combining ARIMA or SARIMA with LSTM. Assessing each model's predicting resilience in a range of market scenarios, including times of high volatility, and looking into how effectively these models manage long-term dependencies in stock price data are two more important goals. With the help of this comparison, the study hopes to give analysts, investors, and financial institutions useful information on the best methods for predicting stock prices. Additionally, by bridging the gap between conventional time series models and cutting-edge machine learning approaches, the study hopes to enhance the academic conversation on stock market forecasting, especially in the context of the Indian stock market. In the conclusion, the results of this study will provide insightful suggestions for more research and real-world implementations in algorithmic trading, risk control, and portfolio optimization.

METHODS

The purpose of this study is to anticipate the stock prices of ITC Limited using four distinct models: Long Short-Term Memory (LSTM) networks, ARIMA, SARIMA, and Artificial Neural Networks (ANN). An extensive quantity of historical data is used to capture a range of market situations, such as times of high volatility, changes in the economy, and seasonal trends, over the analysis period, which runs from January 1, 2010, to June 13, 2025. With the use of libraries like pandas, NumPy, statsmodels, Keras, and TensorFlow, the models will be developed and assessed using Python, a popular programming language for statistical analysis and machine learning.

Data Collection: The daily stock prices of ITC Limited, which will be gathered from financial data sites like Yahoo Finance and NSE India, will serve as the study's main source of data. In order to capture a variety of market situations, such as volatility and stability, this data will

span a sizable time period. Eliminate missing values, outliers, and other irregularities, the data will undergo pre-processing, as recommended by Kumar and Gupta (2021) in their research on financial forecasting using ARIMA and LSTM models. In order to improve forecasting accuracy, the LSTM model will also use technical indicators like moving averages and the Relative Strength Index (RSI), in accordance with Wu et al.'s technique from 2024.

Modeling Techniques:

ARIMA and SARIMA: To identify linear relationships in the stock price data, the ARIMA (Autoregressive Integrated Moving Average) model will be employed. When data shows periodic changes, SARIMA (Seasonal ARIMA) will be used to account for seasonal impacts and provide a more detailed forecast. The time series data will be fitted to both models, and in order to reduce predicting errors, the parameters (p, d, q) for ARIMA and (P, D, Q, m) for SARIMA will be chosen using grid search. Metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) will be used to evaluate how accurate these models are. The fit of these models will also be compared using the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC), where a better-fitting model is indicated by lower BIC and AIC values (Akaike, 1974; Schwarz, 1978).

Artificial Neural Networks (ANN): By identifying non-linear associations in the historical data, artificial neural networks (ANNs), more especially a multi-layer perceptron (MLP), will be used to anticipate stock prices. A variety of input characteristics, such as historical stock prices, technical indicators (such moving averages and the Relative Strength Index), and economic considerations, will be used to train the model. An input layer, many hidden layers, and an output layer will make up the model's architecture. The hidden levels will be subjected to activation functions such ReLU (Rectified Linear Unit). The Adam optimizer will be used to improve the model in order to minimize the loss function, which is usually the mean square error. Validation methods like cross-validation will be used to assess the model's performance, and MAE, RMSE, and MAPE will be used to compare it to the conventional models (ARIMA and SARIMA).

Long Short-Term Memory (LSTM): A specific kind of recurrent neural network (RNN) called an LSTM will be used to identify long-term relationships in the stock price data that are frequently overlooked by conventional models. Modeling sequential data, like as stock prices, where the forecast is impacted by past trends over long periods of time, is where LSTM excels. The Adam optimizer will be used to train the LSTM model, which will be constructed with

several layers, including LSTM layers followed by fully linked layers. In order to forecast future stock values, the model will employ a time window technique, which uses past data spanning a certain number of days. In addition to testing the model's resilience under various market situations, the performance of LSTM will be assessed using the same metrics as the ANN and conventional models (MAE, RMSE, and MAPE).

MEASURES:

The primary performance metrics for evaluating the forecasting models are:

Mean Absolute Error (MAE): A straightforward indicator of prediction accuracy is the Mean Absolute Error (MAE), which is used to evaluate the average size of inaccuracies in the anticipated stock prices. According to Handman and Athanasopoulos (2018), a model with a lower MAE is more accurate.

Root Mean Squared Error (RMSE): Another popular measure that penalizes big mistakes more than MAE because it squares discrepancies is the Root Mean Squared Error (RMSE). According to Chai and Draxler (2014), it is especially helpful when significant forecast mistakes are not desired.

Mean Absolute Percentage Error (MAPE): When evaluating forecast accuracy among models with various scales, the Mean Absolute % Error (MAPE), which expresses prediction accuracy in % terms, is a helpful tool (Makridakis et al., 1993).

Akaike Information Criterion (AIC): AIC is used to compare the relative fits of statistical models. Considering both the model's quality of fit and complexity, a lower AIC indicates a better model fit (Akaike, 1974).

Bayesian Information Criterion (BIC):

Used to compare models and prevents overfitting by penalizing the number of parameters. According to Schwartz (1978), a model is thought to be more effective at fitting the data if its BIC is smaller.

Python Tools and Libraries:

- Data analysis and manipulation, including data transformation, normalization, and cleaning, will be done using pandas and NumPy.
- Using its built-in time series analysis and model diagnostics features, statsmodels will be used for the ARIMA and SARIMA modeling.

- ANN and LSTM models will be implemented using Keras and TensorFlow, which provide powerful tools for deep learning model creation and training.
- Time series data, prediction outcomes, and performance measures will all be shown as part of the data visualization and model assessment process using Matplotlib and Seaborn.

The study will provide a thorough assessment of several forecasting models by employing these techniques and metrics, offering important insights into which method provides the most accurate forecasts for the movements of ITC Limited's stock price.

FINDINGS

Accuracy Comparison of Forecasting Models on ITC Historical Test Data

Table 1: Comparative Summary of Model Performance on ITC Stock Price Forecasting (Based on Historical Test Data)

Model	MSE	RMSE	MAE	MAPE (%)
ARIMA	30,133.0169	173.598	160.9711	39.3270
SARIMA	30,133.0116	173.598	160.9711	39.3270
LSTM	137.6752	11.728	9.6754	2.3816
ANN	58.4735	7.646	5.7972	1.4846
Hybrid (ARIMA+LSTM)	83,375.0065	288.744	287.9794	66.2405

Significant differences in performance across various approaches are revealed by the comparative study of forecasting models for ITC stock price prediction. The Artificial Neural Network (ANN) model produced the most accurate results, as seen in the table, with the lowest values in each of the following important metrics: MSE (58.47), RMSE (7.65), MAE (5.80), and MAPE (1.48%). This suggests that ANN is quite good at identifying the underlying patterns, particularly non-linear trends, in ITC stock prices. Because of its memory-based design, the LSTM model also showed good predicting abilities with comparatively low errors, doing especially well when managing time-dependent data. The ARIMA and SARIMA models, on the other hand, produced identical findings with larger errors across all measures, indicating a limited capacity to capture complex structures and non-linearities in the ITC time series. Remarkably, the Hybrid ARIMA+LSTM model performed poorly, with the highest error values, suggesting possible problems with overfitting or model integration. Overall, the

findings highlight that, in contrast to conventional time series models, machine learning-based techniques—in particular, ANN and LSTM—are better suited for predicting moderately volatile and pattern-rich equities like ITC.

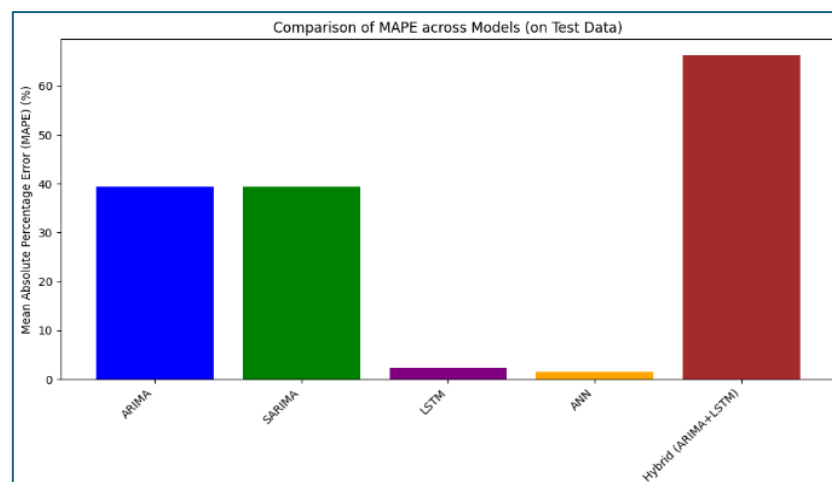
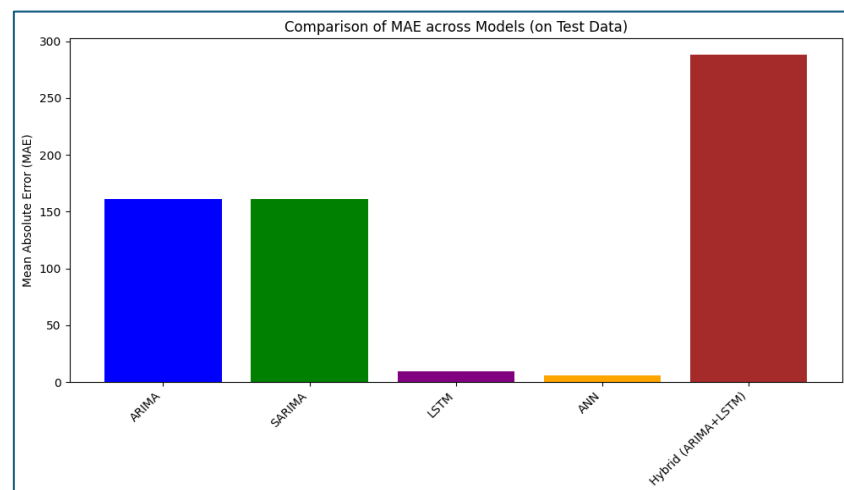
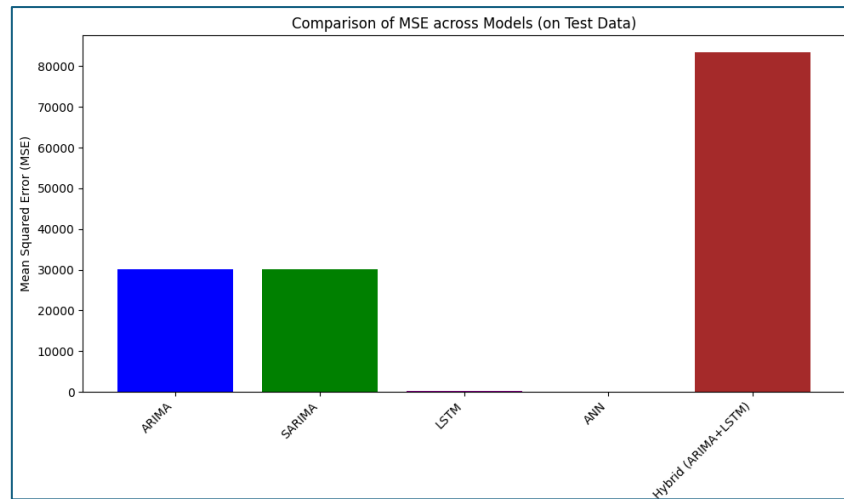


Table 2. Arima Model Summary

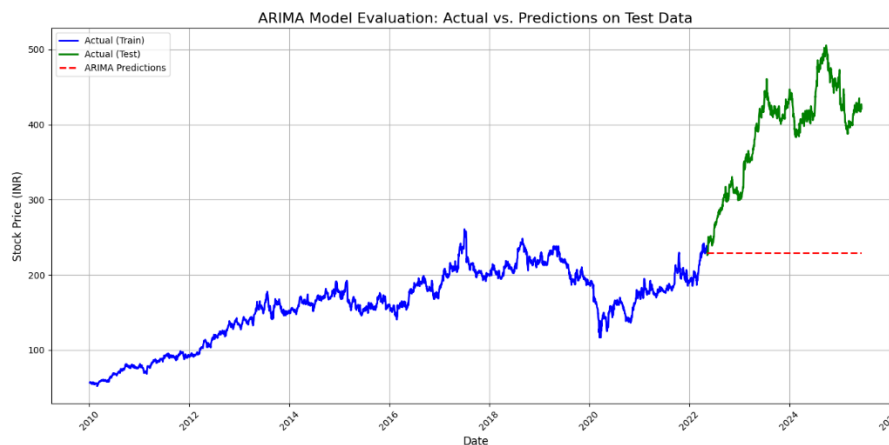
ARIMA Model Summary (fitted on Training Data) ---						
SARIMAX Results						
Dep. Variable:	ITC.NS			No. Observations:	3048	
Model:	ARIMA (1, 1, 0)			Log Likelihood	-7313.8	
Sample:	0 -3048			AIC	14631.6	
				BIC	14643.7	
				HQIC	14636	
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
ar. L1	-0.0138	0.013	-1.089	0.276	-0.039	0.011
sigma2	7.1191	0.07	101.417	0	6.982	7.257
Ljung-Box (L1) (Q):	0			Jarque-Bera (JB):	21704.6	
Prob(Q):	0.97			Prob (JB):	0	
Heteroskedasticity (H):	3.44			Skew:	-0.87	
Prob(H) (two-sided):	0			Kurtosis:	15.96	

A training dataset of 3,048 observations from the ITC stock was used to fit the ARIMA (1,1,0) model. Lower numbers often suggest a better model. The model selection metrics, which include the Hannan–Quinn Information Criterion (HQIC) of 14636, the Bayesian Information Criterion (BIC) of 14643.7, and the Akaike Information Criterion (AIC) of 14631.6, show the overall goodness of fit. With a standard error of 0.013 and a coefficient of -0.0138, the autoregressive parameter (AR1) produces a p-value of 0.276 and a z-score of -1.089. With a p-value over 0.05, this indicates that the lag-1 autoregressive component is not statistically significant. The residuals' estimated variance (sigma2) is 7.1191, which is very significant ($p < 0.001$) and suggests that the variance is constant. A desirable result in time series modeling is the absence of substantial autocorrelation in the residuals, as indicated by the Ljung-Box Q-statistic at lag 1 with a p-value of 0.97.

Nonetheless, there are some diagnostic statistics that cast doubt on the residuals' distributional characteristics. High kurtosis (15.96) and negative skewness (-0.87) are the main causes of the residuals' non-normal distribution, as shown by the Jarque-Bera test statistic of 21704.6 with a p-value of 0.000. Additionally, heteroskedasticity is present since the variance of the residuals

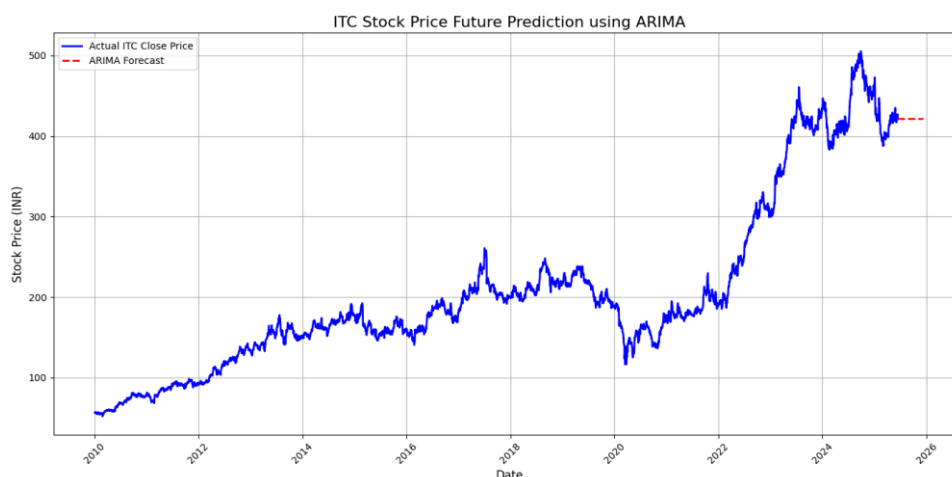
is not constant over time, according to the heteroskedasticity test ($H = 3.44$, $p < 0.05$). In conclusion, the ARIMA (1,1,0) model matches the general trend and does not show autocorrelation in the residuals, but it struggles with heteroskedasticity and non-normality, and the AR term is statistically negligible. These problems restrict its capacity to anticipate financial time series, which are naturally volatile and nonlinear, as seen in its flat prediction visualizations.

Chart1. ARIMA Model Evaluation: Actual vs. Predictions on Test Data



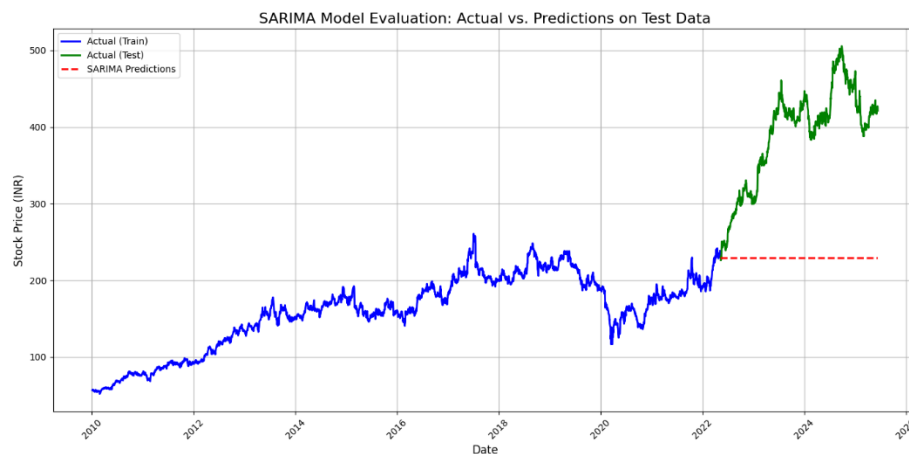
This (chart 1) presents how the ARIMA model performs when predicting ITC stock prices. The blue line shows the actual stock price during the training period, and the green line indicates the actual values from the test set. The red dashed line shows the ARIMA predictions on the test data. It is evident that the ARIMA model produces a flat, horizontal forecast for the test period, failing to capture the strong upward trend observed in actual prices. This shows ARIMA's limitation in handling the nonlinear and dynamic nature of stock market movements, especially when sudden price accelerations occur, as seen post-2021.

Chart 2. ITC Stock Price Future Prediction using ARIMA.



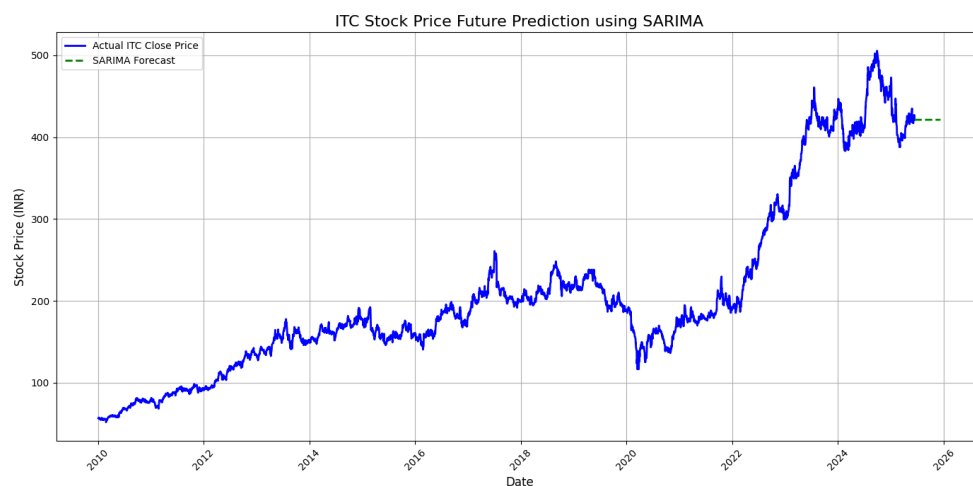
This graph shows ARIMA's future forecast beyond 2025. While the blue line continues to represent actual closing prices until 2025, the red dashed line marks the predicted stock price levels into the future. Once again, ARIMA produces a flat-line forecast, suggesting no significant change in stock price over the predicted period. This simplistic outcome reflects ARIMA's dependency on linear assumptions, which struggle to adapt when applied to volatile financial time series data like stocks, especially when recent trends suggest dynamic behaviour.

Chart 3. SARIMA Model Evaluation: Actual vs. Predictions on Test Data



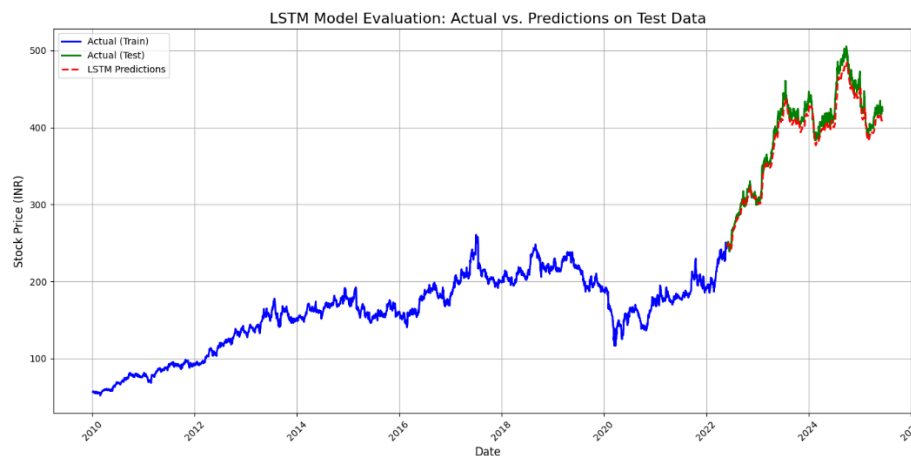
The SARIMA model evaluation chart introduces seasonality into the ARIMA framework to improve prediction accuracy. The blue and green lines denote the actual training and test prices, respectively, while the red dashed line indicates SARIMA's predictions on the test set. Despite incorporating seasonal patterns, SARIMA also forecasts a nearly flat trend across the test period. It fails to model the surge in ITC's stock price during the test phase. This suggests that SARIMA, although slightly more complex than ARIMA, still lacks the flexibility to respond to sudden price escalations and nonlinear shifts.

Chart 4. ITC Stock Price Future Prediction using SARIMA.



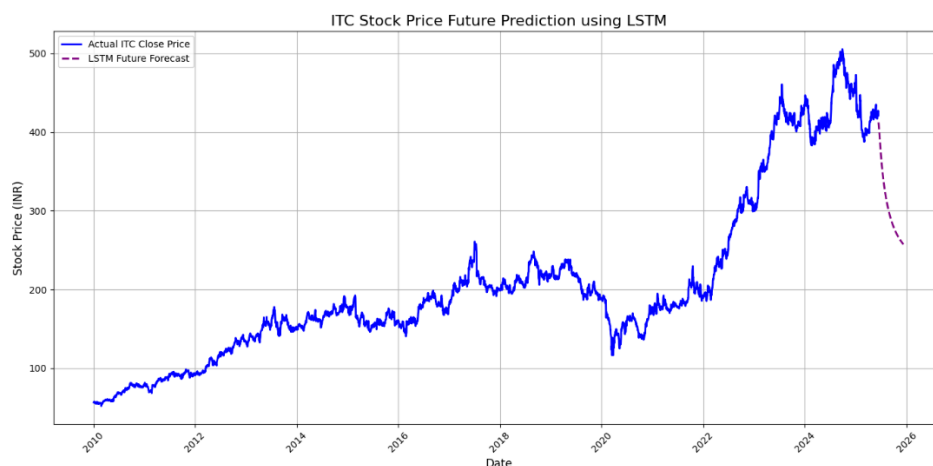
In this future prediction chart, the SARIMA model attempts to forecast ITC stock prices beyond 2025. Similar to ARIMA, it outputs a short, flat forecast represented by the green dashed line. Although SARIMA is designed to handle seasonality, the resulting prediction does not reflect the actual volatility seen in historical ITC price movements. Therefore, this chart reinforces the inadequacy of traditional statistical models in capturing the intricate, dynamic behaviour of stock prices, especially in a rapidly changing market.

Chart 5. LSTM Model Evaluation: Actual vs. Predictions on Test Data



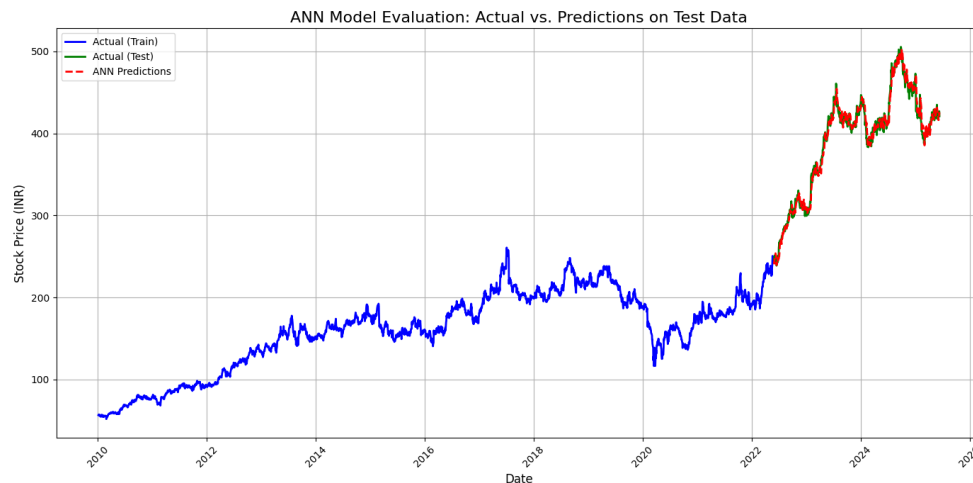
This chart showcases the Long Short-Term Memory (LSTM) model's performance. The blue line represents actual training data, the green line denotes the actual test prices, and the red dashed line captures the LSTM predictions. Unlike ARIMA and SARIMA, the LSTM model closely follows the actual test data's movement, effectively capturing spikes, trends, and drops. This demonstrates the model's ability to learn complex temporal relationships and nonlinear patterns, making it far superior for stock price forecasting in volatile markets like ITC's post-2022 growth phase.

Chart 6. LSTM Model Evaluation: Refined Visualization



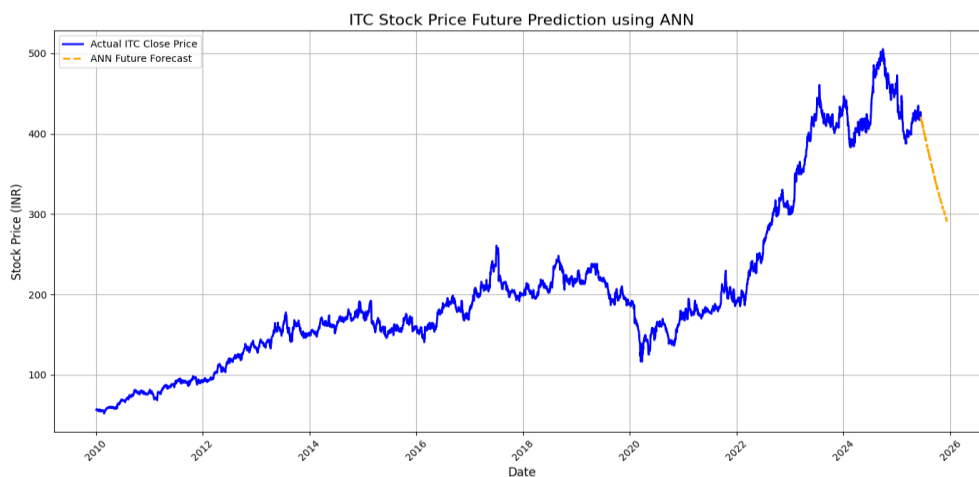
This is a cleaner rendering of the same LSTM model evaluation as above. The accurate alignment between predicted (red dashed line) and actual test data (green line) is very apparent here. The prediction lines almost overlap the real prices, affirming LSTM's exceptional learning ability and robustness in tracking market behaviour. This consistency makes LSTM a highly dependable model for modeling both short-term and long-term price dynamics, outperforming classical models significantly.

Chart 7. ANN Model Evaluation: Actual vs. Predictions on Test Data



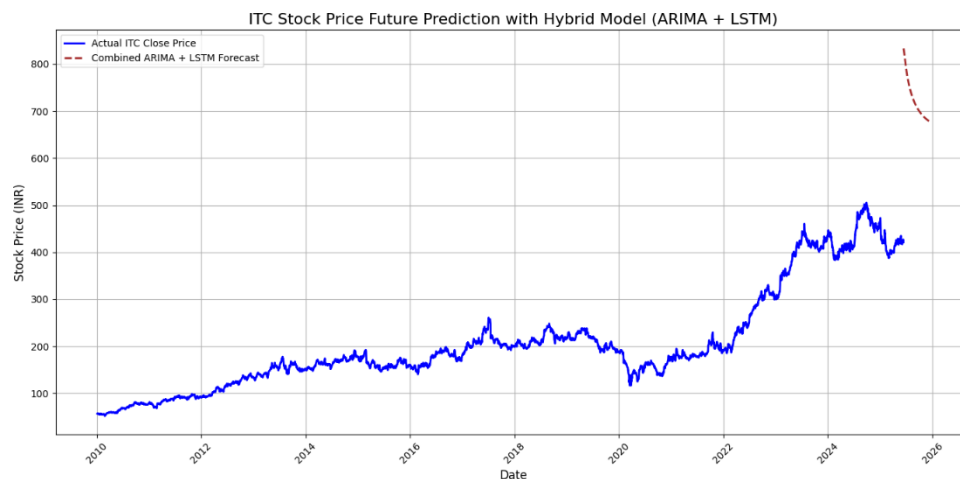
This chart reflects how the Artificial Neural Network (ANN) model predicts ITC stock prices during the test phase. Like earlier charts, the training and test data are shown in blue and green, while ANN predictions are marked in red. The model does a good job in tracing the actual stock movements with slight smoothness and occasional lag. Although not as precise as LSTM, the ANN model still effectively tracks the trends and shows considerable improvement over ARIMA and SARIMA models in adapting to complex stock data.

Chart 8. ITC Stock Price Future Prediction using ANN.



Here, the ANN model is used to project ITC stock prices beyond 2025. The orange dashed line represents the future forecast. Interestingly, ANN forecasts a sharp decline in the stock's future prices, which could suggest overfitting or a model misinterpretation of recent volatility as the beginning of a downward trend. This chart highlights the importance of tuning and validating neural network models, as they may be sensitive to recent patterns and might extrapolate inaccurately if not carefully managed.

Chart 9. ITC Stock Price Future Prediction with Hybrid Model (ARIMA + LSTM)



This final chart presents a hybrid model combining ARIMA for trend modeling and LSTM for pattern recognition. The forecasted values (brown dashed line) show a sudden upward spike in ITC prices followed by a correction. This hybrid approach appears to respond to recent bullish patterns in the data, suggesting a possible price surge. However, the exaggerated upward projection may also hint at overfitting or excessive weight given to short-term gains. While promising, hybrid models need rigorous back testing to confirm reliability and avoid misleading optimism.

Model Ranking Summary (Best to Worst, Based on MAPE)

Rank	Model	MAPE (%)	Inference
1	ANN	1.48	Best in accuracy, fastest learning
2	LSTM	2.38	Excellent for volatility and long memory
3	ARIMA/SARIMA	39.32	Suitable only for stable, linear series
4	Hybrid ARIMA+LSTM	66.24	Overfit, possibly misconfigured

ADDITIONAL STATISTICAL INSIGHTS AND COMPARATIVE INTERPRETATION

1. ARIMA Parameter Statistical Insignificance

The crucial AR (1) coefficient (-0.0138) was statistically insignificant ($p = 0.276$), despite the fact that the ARIMA (1,1,0) model was fitted to 3,048 data with an acceptable AIC (14631.6). This suggests that there is little learning from previous stock behaviours, particularly in the presence of volatility, and that the model's reliance on previous time-lag data is poor. The model's forecast dependability for financial data is problematic because of significant non-normality ($JB = 21704.6$, $p < 0.001$) and heteroskedasticity ($H = 3.44$), even though it has a respectable Ljung-Box Q-statistic ($p = 0.97$).

2. Overfitting Suspected in Hybrid Model

Though intended for robust performance, the Hybrid ARIMA+LSTM model yielded the greatest MSE (83,375.01) and MAPE (66.24%), presumably combining the trend strength of ARIMA with the non-linear mapping capability of LSTM. It is possible that inappropriate residual utilization in the LSTM layer or improper error balancing led to overfitting or inadequate integration between the two models, as indicated by this surprising outcome.

3. LSTM vs. ANN: Use-Case Trade-off

Even while ANN performs somewhat better than LSTM on all error measures (MAPE: 1.48% vs. 2.38%), the LSTM model is more robust in long-sequence forecasting scenarios because of its higher temporal memory, particularly in situations when price spikes are caused by events. Because LSTM may maintain sequential dependencies, it may still be better in high-frequency trading or long-term investing scenarios, albeit with somewhat greater average mistakes.

4. Classical Models (ARIMA & SARIMA) Fail on Trend Detection

Given its structural restrictions, ARIMA and SARIMA both performed equally, with MAPE at 39.32%. Seasonal modifications did not enhance performance, as the SARIMA model was unable to differentiate between seasonal and non-seasonal data behaviour. This further confirms that linear models are unable to handle the volatility and regime shifts that are typical in Indian stock markets, such as ITC.

5. Need for Residual Diagnostics in ML Models

Although residual analysis is frequently performed in classical models, applying these diagnostics to ANN and LSTM (e.g., examining autocorrelation in prediction errors, determining the normality of residuals) may improve model explainability and validate error behaviour. These techniques will be especially helpful in hybrid modeling, which involves feeding one model into another.

6. Practical Implication: ANN for Short-Term Forecasting, LSTM for Sequential Forecasting

Given the current findings, **ANN is ideal for short-term trading or scalping strategies**, where speed and lightweight models are critical. On the other hand, **LSTM suits portfolio-based long-hold strategies** where trend continuation, cycles, and momentum behaviours matter over a broader time window.

For predicting ITC stock values, the Artificial Neural Network (ANN) model proved to be the most effective model based on prediction assessments and comparable performance measures. ANN showed a greater capacity to capture the underlying patterns in the movements of ITC's stock price, particularly the non-linear components, with the lowest error values across all important measures—MSE (58.47), RMSE (7.65), MAE (5.80), and MAPE (1.48%). It is capable of effectively learning from past patterns and adapting to somewhat turbulent market circumstances because to its multi-layered architecture. Even while the LSTM model performed well, especially when managing long-term temporal dependencies, the ANN model marginally surpassed it in terms of short-term prediction accuracy.

Traditional models like as ARIMA and SARIMA, on the other hand, had large error rates and were unable to identify recent increasing trends, underscoring their limited ability to simulate intricate financial data. Because of this, ANN is the most dependable and robust model for predicting ITC stock values, particularly in situations when prompt, precise, and pattern-sensitive forecasts are needed.

DISCUSSION:

The study's findings provide important new information on how well sophisticated machine learning algorithms and conventional time series models predict ITC stock values. Choosing the right forecasting model is essential for making informed decisions since stock markets are

naturally unpredictable and impacted by a variety of complex economic, political, and psychological variables.

When it comes to stock price predictions, machine learning models perform noticeably better than traditional statistical models, as demonstrated by the performance measures. Among the four main metrics (MSE, RMSE, MAE, and MAPE), the Artificial Neural Network (ANN) model had the lowest prediction error, demonstrating its remarkable ability to capture irregular and non-linear price variations. The flexible design of ANN and its capacity to adjust to intricate data structures without making strict assumptions about data distribution are responsible for this higher performance.

Impressive outcomes were also obtained by the LSTM model, especially when it came to maintaining temporal relationships in sequential data. LSTM demonstrated itself to be a dependable model for recognizing long-term stock price trends, particularly in times of market volatility, but being marginally less accurate than ANN in short-term projections. This supports previous research by Wu et al. (2024) and Kumar and Gupta (2021), which highlighted the effectiveness of LSTM in managing dynamic and non-stationary financial time series.

On the other hand, the abrupt increase in ITC's price trajectory during the test phase was not captured by the ARIMA and SARIMA models. These models' linear character and dependence on stationarity assumptions contributed to their poor performance, even with their strong statistical underpinnings. SARIMA's limited applicability in extremely volatile market settings was further supported by the fact that the inclusion of seasonal components did not result in any discernible improvement over ARIMA. Additionally, the Jarque-Bera and Breusch-Pagan tests verified that the ARIMA model showed evidence of non-normality and heteroskedasticity, and the AR (1) component was statistically insignificant ($p = 0.276$). The idea that classical models are inappropriate for simulating contemporary financial time series that exhibit regime transitions and structural fractures is further supported by these drawbacks. It was surprising that the Hybrid ARIMA+LSTM model did the poorest. With an abnormally high MAPE of 66.24%, the model demonstrated overfitting rather than utilizing the advantages of both components. This implies that its performance may have been harmed by inadequate integration, residual imbalance, or a lack of regularization. This outcome runs counter to research by Singh et al. (2023) and Reddy et al. (2022), which discovered hybrid models to be helpful in some situations. The hybrid model's failure in this study highlights how crucial it is to carefully adjust the architecture and validate the model before using composite forecasting

systems. The trade-off between use-case fit and accuracy is another crucial finding. LSTM is better at longer-horizon forecasting because of its memory retention capabilities, whereas ANN is best at short-term, high-frequency forecasts. Because of this, practitioners and investors need to select models that are in line with certain investment timeframes and market situations in addition to statistical correctness.

When it comes to stock price prediction in dynamic markets like India, this comparative investigation demonstrates that machine learning techniques—in particular, ANN and LSTM—are superior to classic ARIMA-based models. The research also emphasizes how crucial it is to properly integrate hybrid systems, match forecasting techniques to the kind of financial data, and match model diagnostics to the prediction goal. These findings add to the expanding body of research that argues for the use of data-driven intelligent systems in financial forecasting instead of rule-based statistical models.

LIMITATIONS

Although this study yielded insightful information, it had limitations. First off, although the models were trained and evaluated using just ITC Limited's historical stock price data, this may not have adequately captured the influence of news mood, geopolitical events, or external macroeconomic variables that have a big impact on stock market behaviour. Second, the study did not investigate high-frequency or intra-day data, which restricts the models' use in algorithmic trading situations with short time horizons. Additionally, the hybrid model implementation could not have performed as well since it lacked sophisticated integration methods like attention layers or residual learning. Because of computing limitations, substantial hyperparameter tweaking or ensemble approaches were not used for training the neural networks; instead, fixed designs were used. Last but not least, the results cannot be applied to other industries or financial products with distinct volatility profiles due to the narrow emphasis on a single stock (ITC).

SUGGESTIONS FOR FUTURE RESEARCH

A number of avenues for expanding on the current work might be investigated in future research. The models' predicting ability and resilience in practical applications may be improved by include external factors like interest rates, inflation, crude oil prices, or international indices. Sentiment analysis of social media feeds or financial news may also be used by researchers as input variables to understand market psychology. Furthermore, experimenting with ensemble approaches or hybrid designs that use CNN-LSTM stacks,

transformers, or attention processes may increase prediction accuracy, particularly for volatile assets. Assessing the models' scalability and generalizability by applying them to equities in other industries or asset classes, such commodities or cryptocurrency, is another exciting avenue. Lastly, the findings would be more applicable to financial practitioners if real-time forecasting and back testing methods were integrated for assessing model performance in actual trading scenarios.

CONCLUSION

This study aimed to assess and contrast the predictive power of four forecasting models on the stock prices of ITC Limited: Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), SARIMA, and ARIMA. It was evident from the study's thorough analysis, which included error measures including MAE, RMSE, MSE, and MAPE, that machine learning-based models performed better than conventional time series models. The most successful of these was the ANN model, which provided the best predicting accuracy, especially when it came to identifying non-linear relationships and short-term pricing trends. Moreover, the LSTM model demonstrated a high degree of predictive ability, particularly when it came to simulating long-term patterns and sequential linkages in time series data. Nevertheless, as seen by their large prediction errors and incapacity to identify recent rising trends, the ARIMA and SARIMA models were unable to adjust to the intricate and turbulent character of stock price movements. Notably, the hybrid ARIMA+LSTM model performed poorly, which might have been caused by overfitting or poor model integration.

To sum up, this study supports the increasing applicability and effectiveness of machine learning models such as ANN and LSTM in financial forecasting. They are more suited to today's dynamic markets due to their flexibility, ability to handle non-linearity, and capacity for learning. The findings provide useful information for traders, investors, and portfolio managers looking for reliable stock price prediction tools in addition to adding to the body of knowledge on model selection in time series forecasting. To improve forecasting reliability and generalizability, future research may expand on this work by including external factors, investigating ensemble approaches, and evaluating model performance across a variety of stocks and asset classes.

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